YAKEEN

NEET 2.0 2027

Physics | Lecture - 2

Vector

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Todays Goal.

Type of vector *

Magnitude of vector → *

Component of vector → *

Question

A. Physical quantity which does not have direction must be scalar.

B. Physical quantity which have direction must be vector.

1. Both are true
2. Both are false
3. A true B false
4. A false B true

(1)

(i) Angle b/w $\vec{A}$ & $\vec{B}$ = 120°

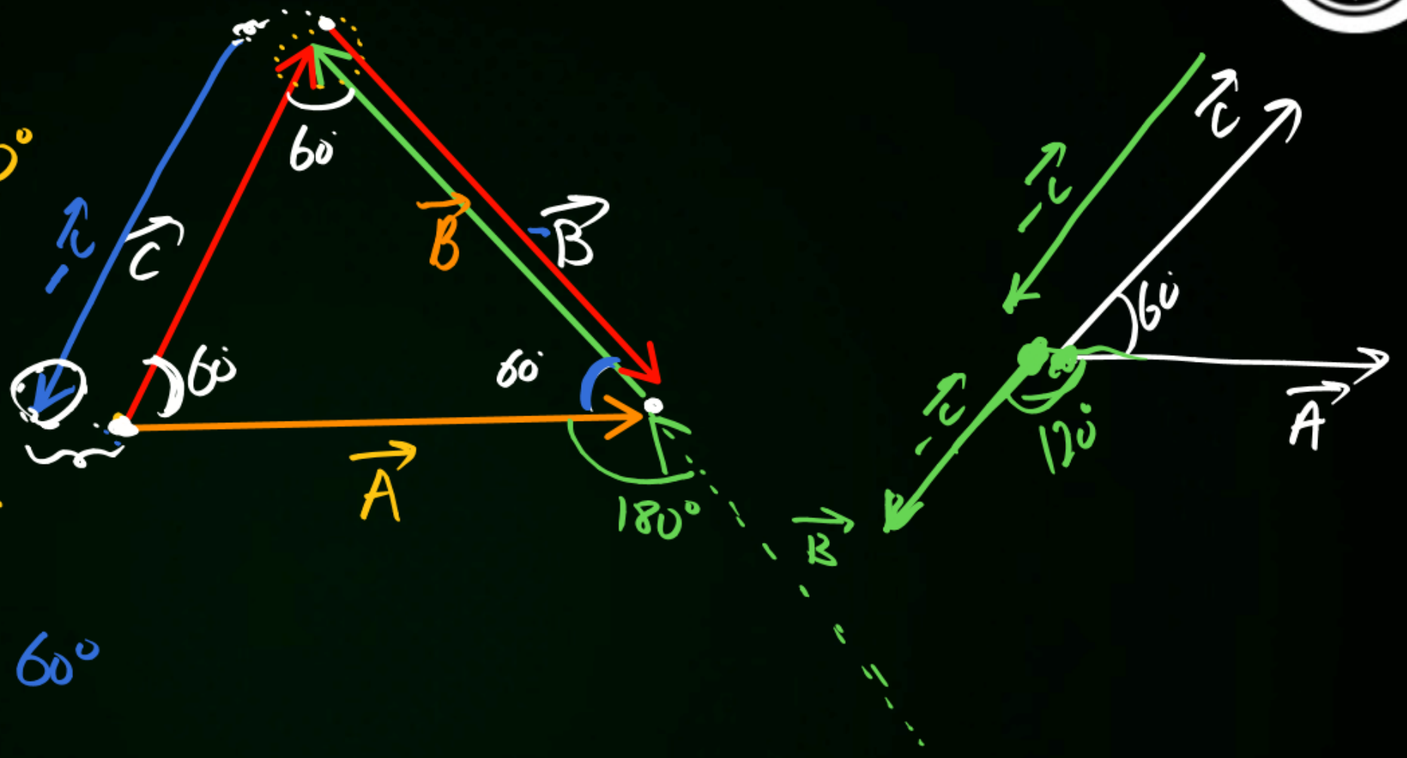
(ii) Angle b/w $\vec{A}$ & $\vec{C}$ = 60°

(iii) Angle b/w $\vec{B}$ & $\vec{C}$ = 60°

(iv) Angle b/w $\vec{A}$ & $-\vec{B}$ = 60°

*(v) Angle b/w $\vec{A}$ & $-\vec{C}$ = 120° = 180 − 60

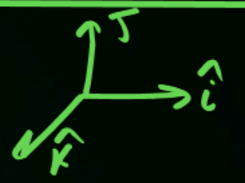
(vi) Angle b/w $-\vec{C}$ & $-\vec{B}$ = 60°



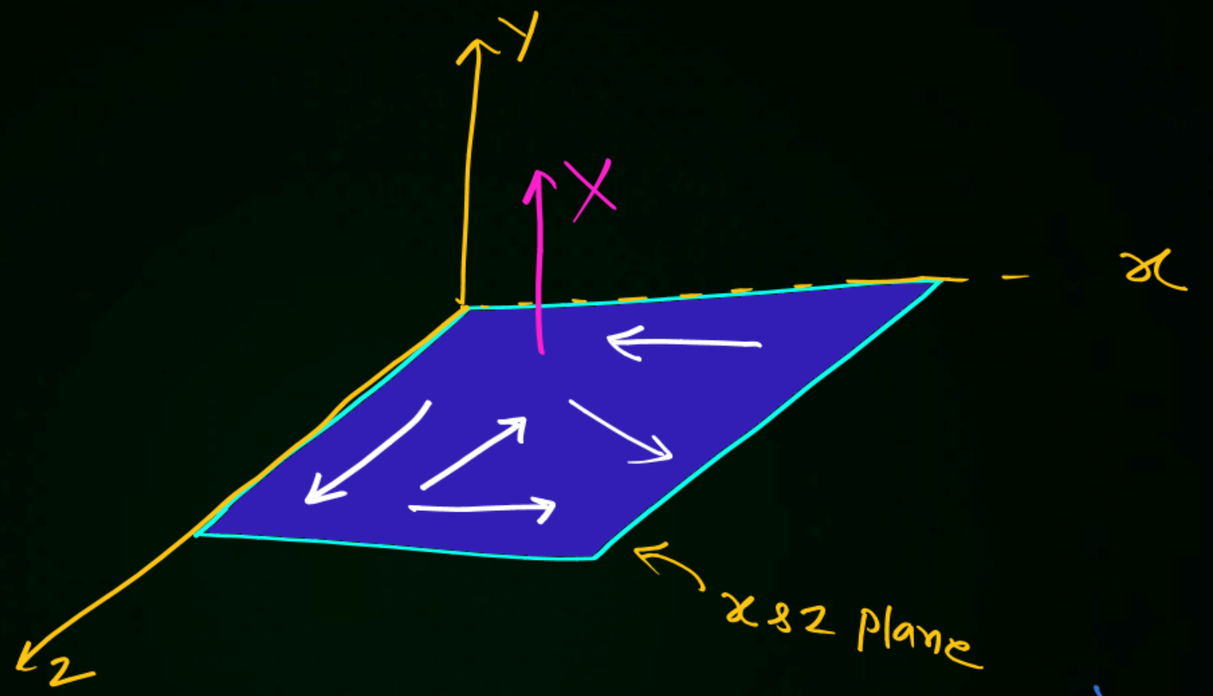
Type of vector

Type of vector	Magnitude	direction. (Angle B/w them)	diagram
(1) equal (Dono Property Same 3 same nature)	Same	Same ($\theta = 0$)	$\vec{A}$ $\vec{B}$
(2) Parallel vector	May or may not same.	Same ($\theta = 0$)	$\vec{B}$ $\vec{A}$
3. Opposite vector	Same	Opposite dir $\theta = 180$	$\vec{A}$ $-\vec{B}$
4. Anti-Parallel	May or may not Same.	* Opposite dir $\theta = 180$.	$\vec{A}$ $\vec{B}$
5. orthogonal vector (⊥r vector)	//	$\theta = 90$.	$\vec{A}$ 90° $\vec{B}$

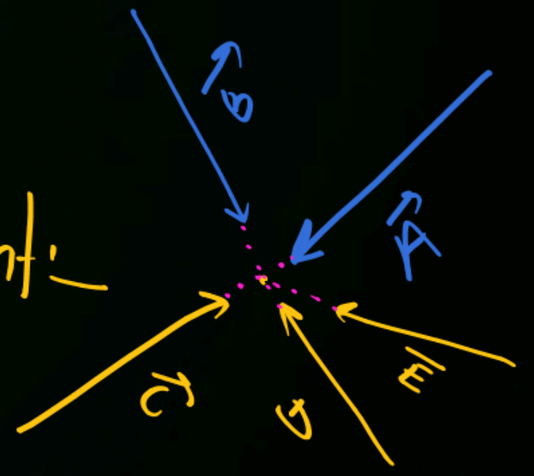
Type of vector

Type of vector	Magnitude	direction. (Angle B/w them)	diagram
6. Unit vector $\vec{F} = 10\hat{i}$ $\hat{F} = \frac{\vec{F}}{\lvert\vec{F}\rvert} = \frac{10\hat{i}}{10} = \hat{i}$	one (1) $\vec{A} = A\hat{A}$ $\hat{A} = \frac{\vec{A}}{A}$	only have dir^n not have Unit.	
7. Null vector	Zero	* dir^r not defined on any arbitrary dir^n angle b/w them is 0° or 180°	 $F_{net} = 0$
# 8. Co-liner vector ↳ vector actiy along Same line	May or may not Same		

9. Coplane Vector → vectors acting in same plane.



(10) Concurrent vectors → Pair of vectors passing through same point or point of action same pa



11. axial vector ⟶ Direction of axial vector along axis of Rotn

Ex – Angular velocity ($\vec{\omega}$) ✓

Angular momentum ($\vec{L}$) ✓

Torque ($\vec{\tau}$) ✓

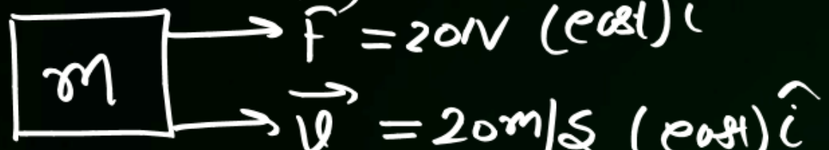
~~M.O.I~~

① all Parallel vectors are equal vector ⟶ Fase

② all equal vectors are Parallel vectors ⟶ True.

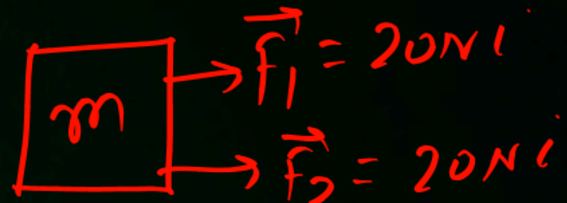
③ all anti-Parall* are opposite vectors ⟶ Fals

④ all opposite vectors are anti-Parall ⟶ Tatue



$\vec{F}$ & $\vec{v}$ are equal vector ✗

$\vec{F} = \vec{v}$ ✗



F_1 & F_2 are equal vector

$\vec{F_1} = \vec{F_2}$

Different Type of vector

(1) Pair of Parallel Vector $(\vec{A},\vec{B})$ $(\vec{B},\vec{C})$ $(\vec{A},\vec{C})$ $(\vec{E},\vec{F})$ ✓

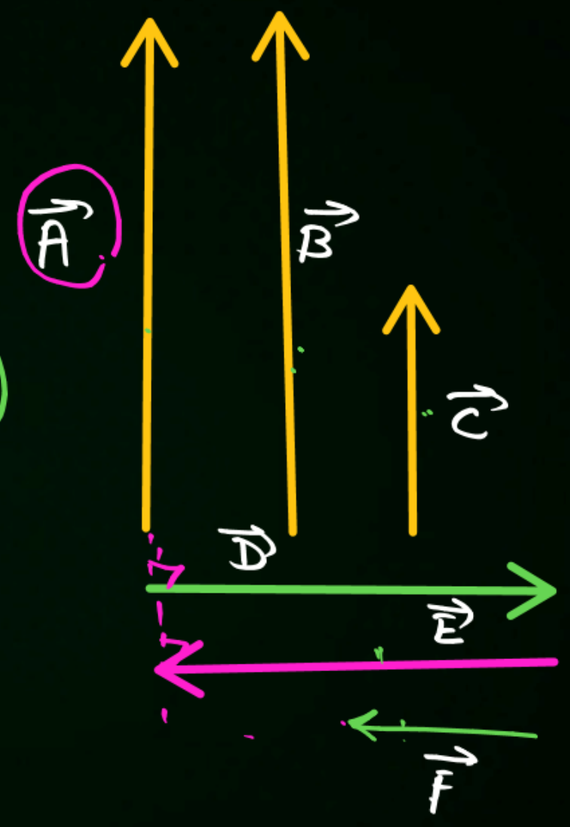
(2) Pair equal vector $(\vec{A},\vec{B})$

(3) Pair of opposite (-ve vector) → (D E)

(4) Anti-Parallel vector → $(\vec{D},\vec{F})$ $(\vec{D},\vec{E})$

(5) orthogonal vector → (A D) (A, E) (A, F)
→ (B D) (B E) (B F)
(C D) (C E) (C F)

#

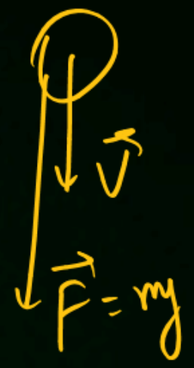


Object Project upward



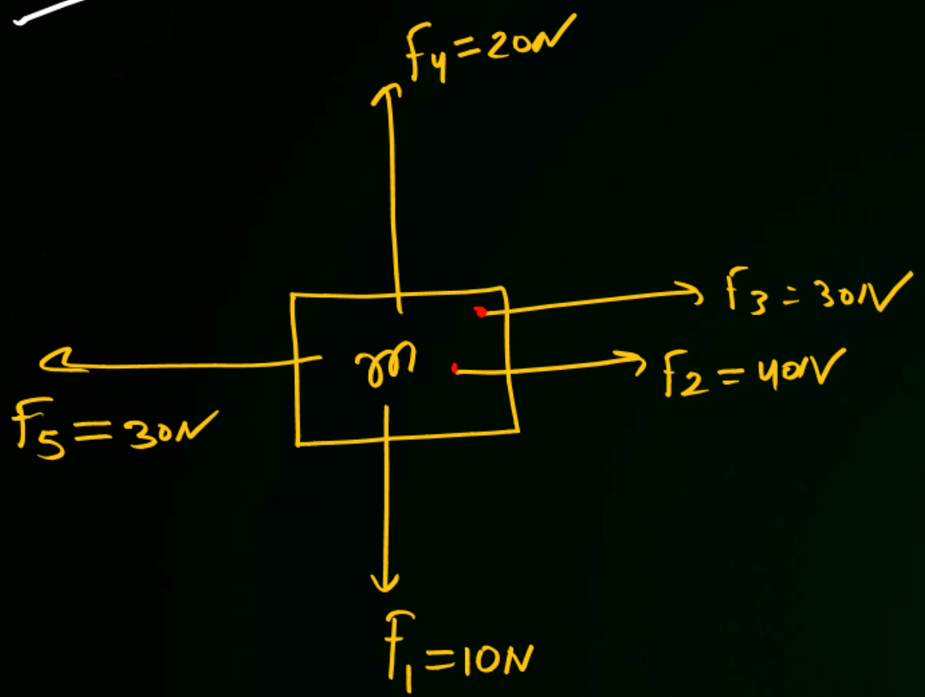
$\vec{V}$ & $\vec{F}$ are anti-Parallel.

Object is thrown downward



$\vec{F}$ & $\vec{V}$ are Parallel

M/W ghar pe solve

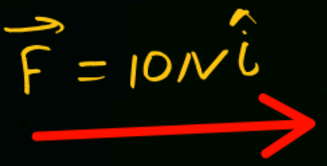


- all $\vec{F_1}$, $\vec{F_2}$, $\vec{F_3}$, $\vec{F_4}$ & $\vec{F_5}$ are coplanar

Pair	Type of vector
f_1 & F_2	
f_1 & f_4	
F_2 & f_5	
f_3 & f_5	
F_2 & F_4	
F_5 & F_3	
F_4 & F_5	

If any vector is multiplied with Positive scalar then its magnitude will change but direction remains same → If scalar is -ve then magnitude change & dirn opposite

$\vec{F} = 10N\hat{i}$



Find $3\vec{F} = 3 \times 10N = 30N\hat{i}$



$3\vec{F}$

(#) $\frac{\vec{F}}{2} = \frac{10N\hat{i}}{2} = 5N\hat{i}$



$\vec{F} = 10N\hat{i}$



$-2\vec{F} = -2 \times (10N\hat{i})$

$= -20N\hat{i}$



$-2\vec{F}$

(8) If $\vec{A} = 30\hat{i} + 20\hat{j}$ then

find (i) $2\vec{A} = 2(30\hat{i} + 20\hat{j}) = 60\hat{i} + 40\hat{j}$

$$\frac{\vec{A}}{2} = 15\hat{i} + 10\hat{j}$$

$$3\vec{A} = 3(30\hat{i} + 20\hat{j})$$

$$-5\vec{A} = -5(30\hat{i} + 20\hat{j})$$

$$= -150\hat{i} - 100\hat{j}$$

Draw given vector with direction.

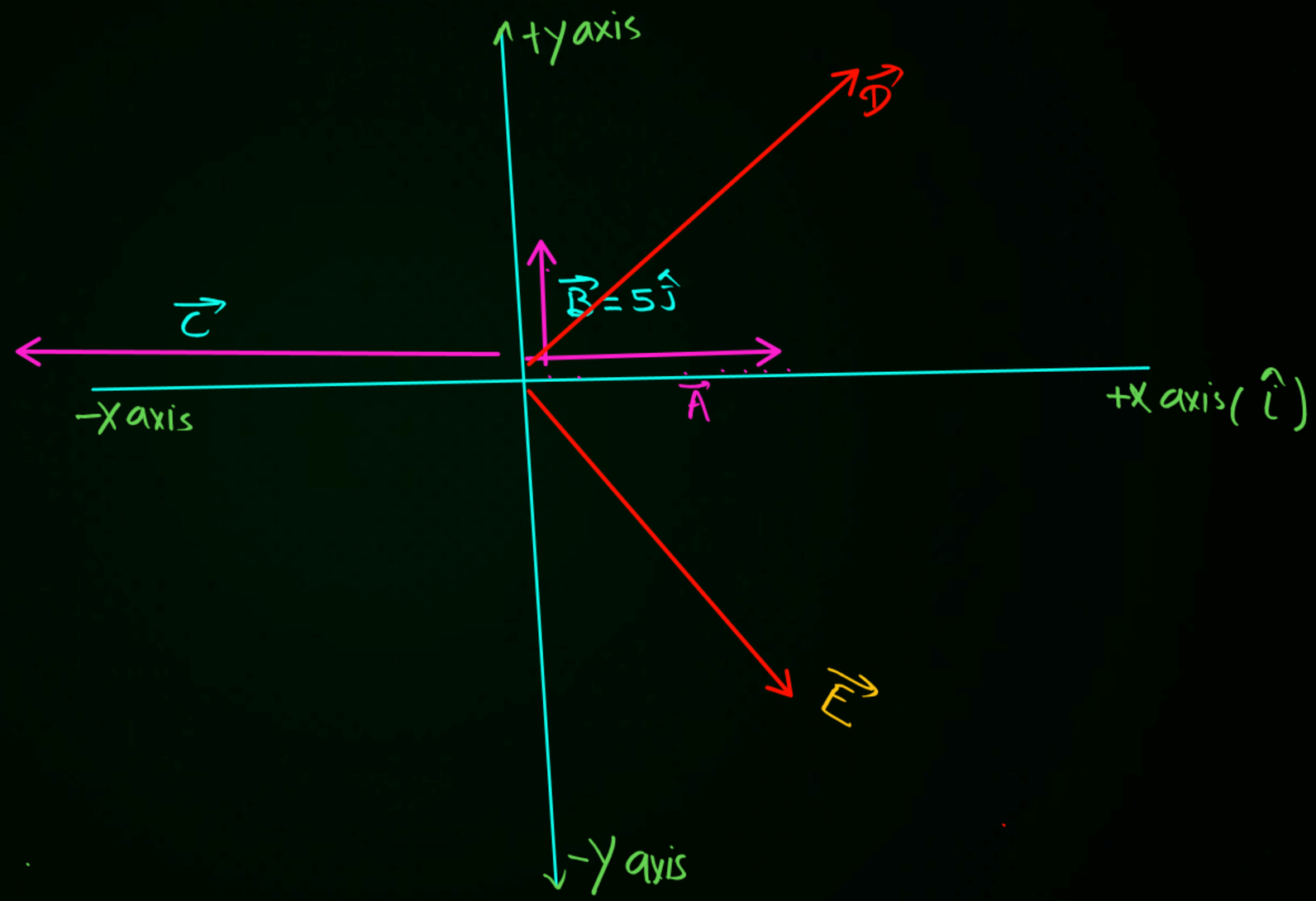
(1) $\vec{A} = 10\hat{i}$

(2) $\vec{B} = 5\hat{j}$

(3) $\vec{C} = -20\hat{i}$

(4) $\vec{D} = 10\hat{i} + 10\hat{j}$

(5) $\vec{E} = 10\hat{i} - 10\hat{j}$



⑥ force $\vec{F}_1 = 30N$ at 30° east of south

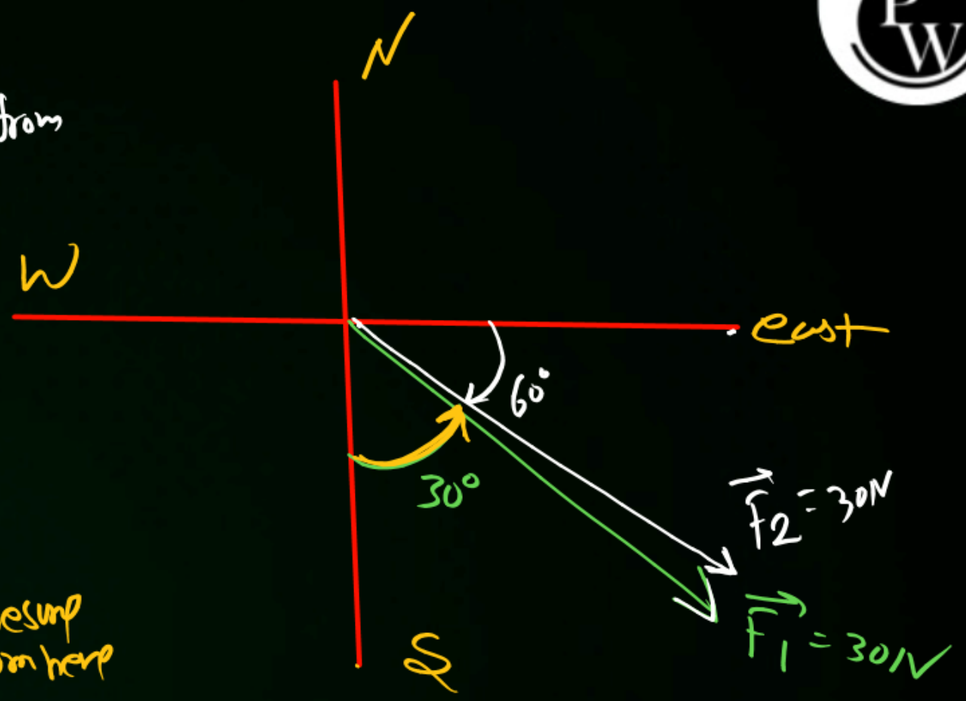
इसे

Angle measure from here

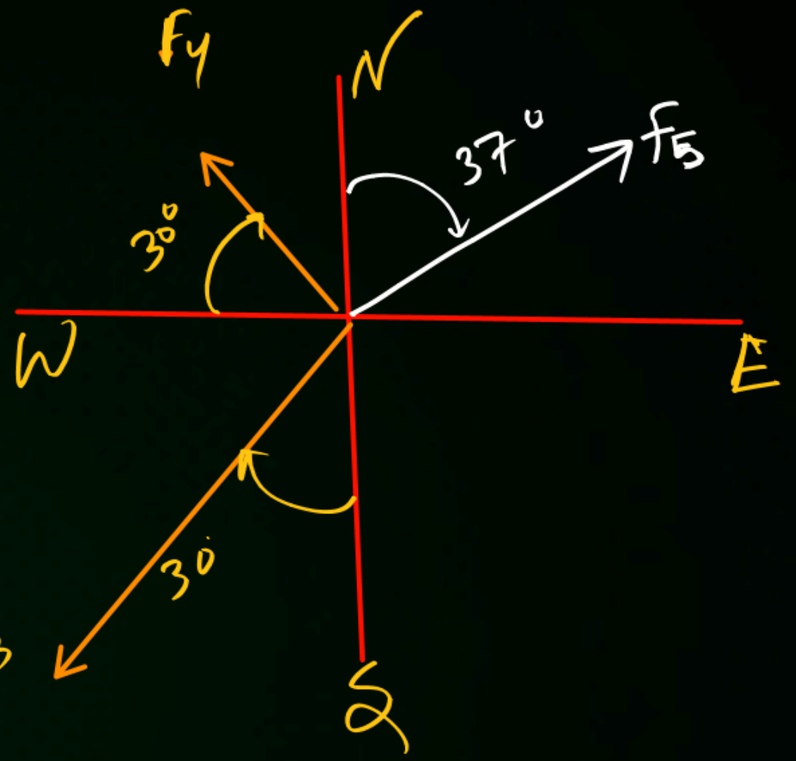
⑦ force $F_2 = 30N$ at 60° south of east

इसे

Angle measure from here



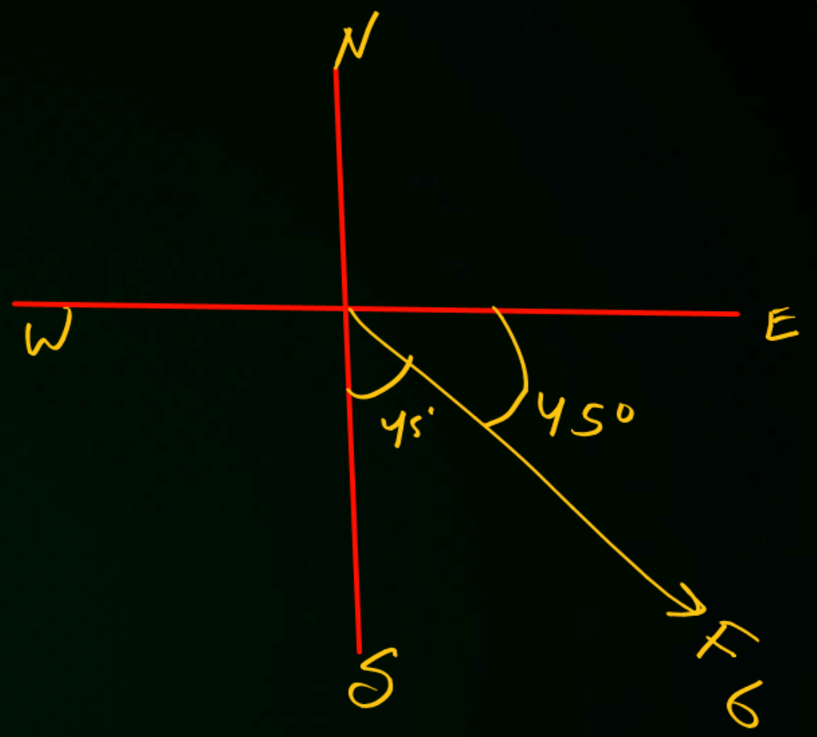
(8) force $\vec{F_3} = 30N$ at 30° west of south.



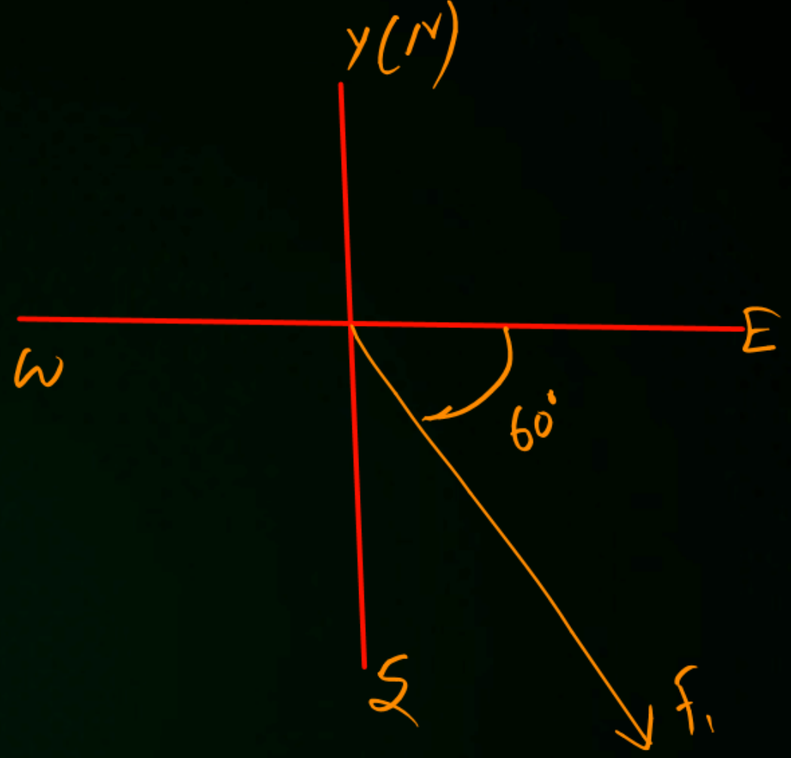
(9) $\vec{F_4} = 15N$ at 30° North of West

(10) $F_5 = 20N$ at 37° East of North

(11) $\vec{F}_6 = 30N$ at south of east

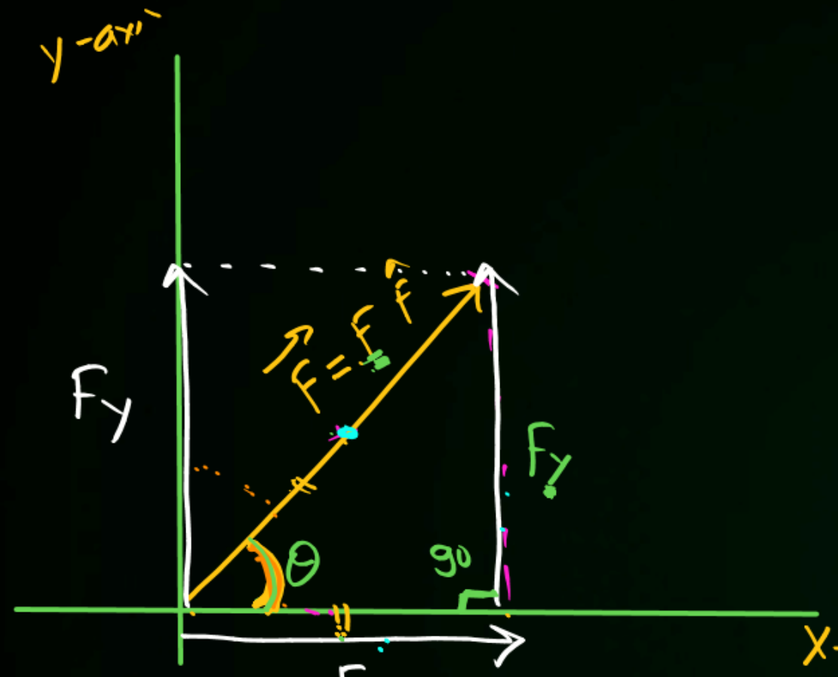


(2) force $\vec{F_1} = 30N$ at east-60°-south.



Component of vector [तोड़ना]

Rectangular



Component of force along X-axis

$$\cos\theta = \frac{F_x}{F} = \frac{\text{Base}}{\text{Hypotenuse}}$$

$$F_x = F\cos\theta \quad \text{—(I)}$$

Comp.

$$\sin\theta = \frac{P}{H} = \frac{F_y}{F}$$

$$F_y = F\sin\theta \quad \text{—(II)}$$

$$\tan\theta = \frac{F_y}{F_x}$$

Angle of $\vec{F}$ from $\vec{F_x}$

Magnitude of vector

$$\vec{F} = F_x\hat{i} + F_y\hat{j}$$

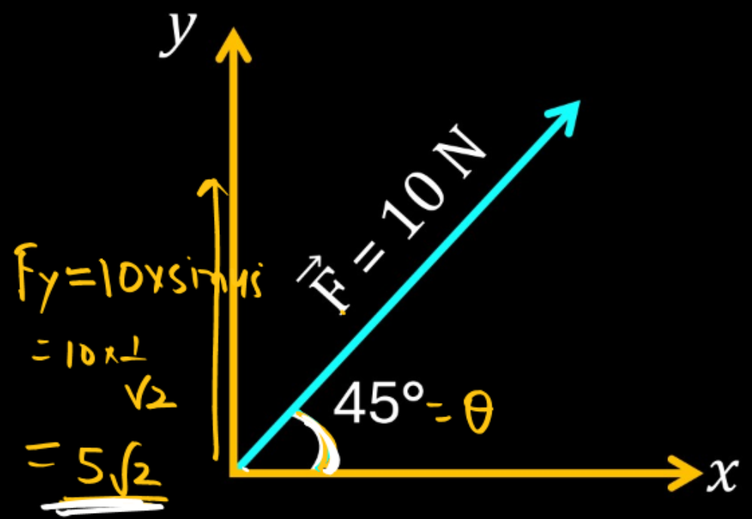
$$(I)^2 + (II)^2$$

$$F^2\cos^2\theta + F^2\sin^2\theta = F_x^2 + F_y^2$$

$$F^2(\sin^2\theta + \cos^2\theta) = F_x^2 + F_y^2$$

$$F = \sqrt{F_x^2 + F_y^2}$$

$\sqrt{2} \times \sqrt{2} = 2$



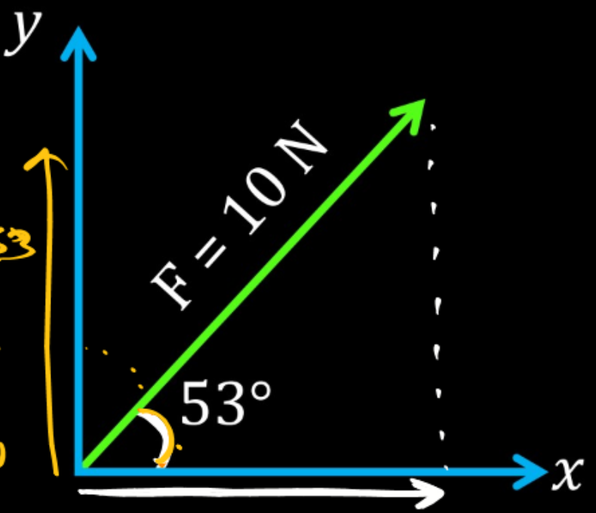
$F_y = 10 \times \sin 45$

$= 10 \times \frac{1}{\sqrt{2}}$

$= 5\sqrt{2}$

$F_x = 10 \cos 45°$

$= 10 \times \frac{1}{\sqrt{2}} = \frac{10}{\sqrt{2}} = \frac{5 \times 2\sqrt{2}}{\sqrt{2}} = 5\sqrt{2}$



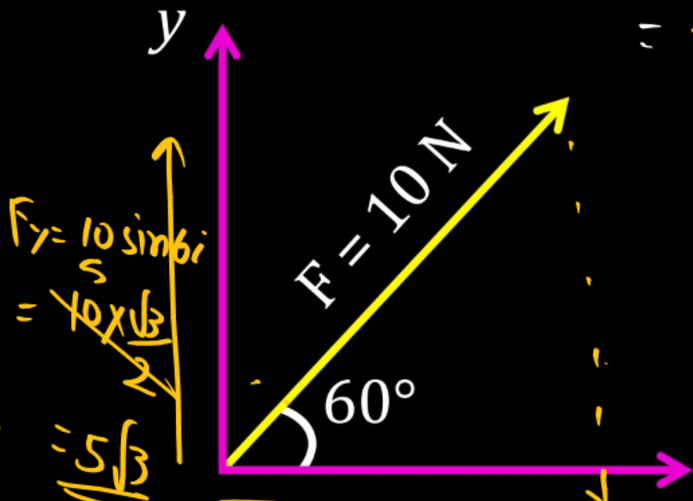
$F_y = 10 \sin 53$

$= \frac{2}{10} \times \frac{4}{5}$

$= 8N\ \hat{j}$

$F_x = 10 \times \cos 53°$

$= \frac{2}{10} \times \frac{3}{5} = 6\hat{i}$

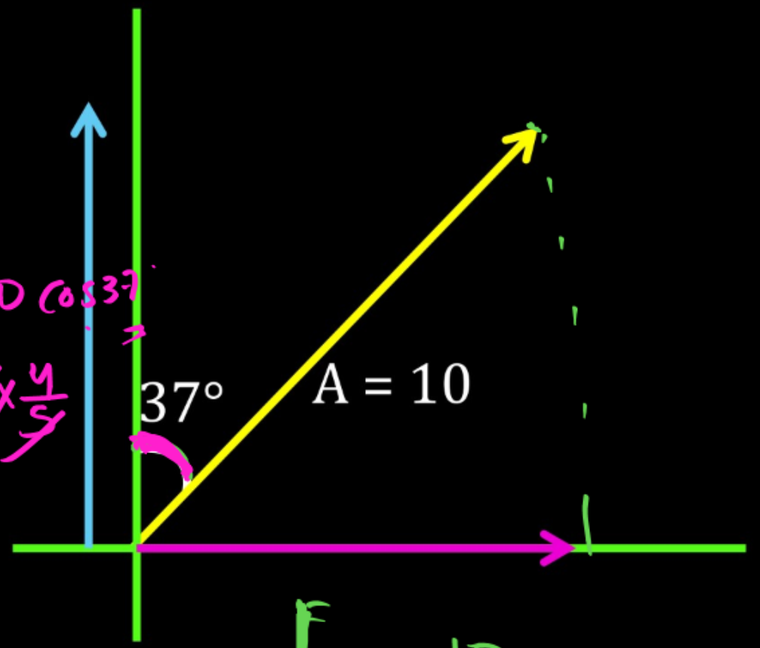


$F_y = 10 \sin 60$

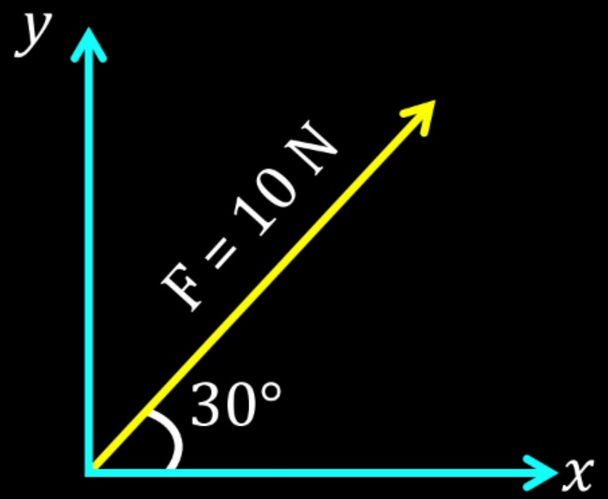
$= 10 \times \frac{\sqrt{3}}{2}$

$= 5\sqrt{3}$

$F_x = 10 \cos 60 = 10 \times \frac{1}{2} = 5$

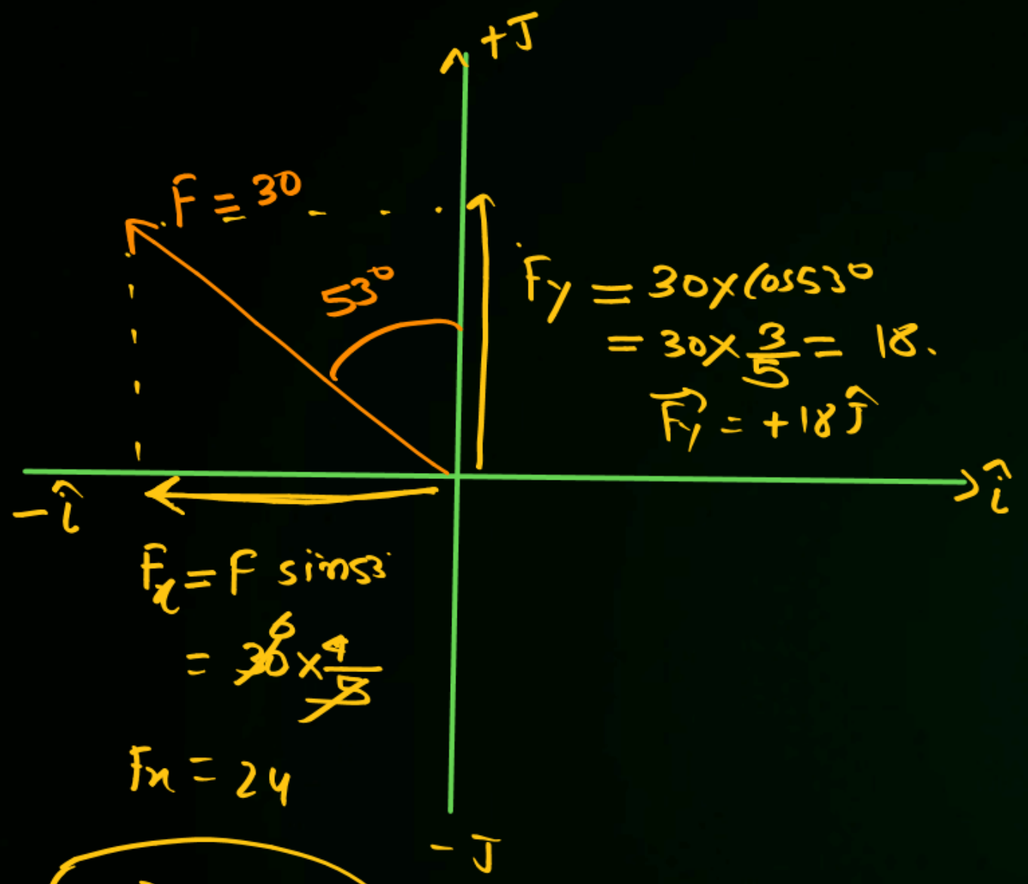


$A_y = 10\cos 37° = 10 \times \frac{4}{5} = 8$



$F_x = 10\cos 37° = 10 \times \frac{4}{5}$

$f_x = 10\sin 37 = 10 \times \frac{3}{5} = 6$



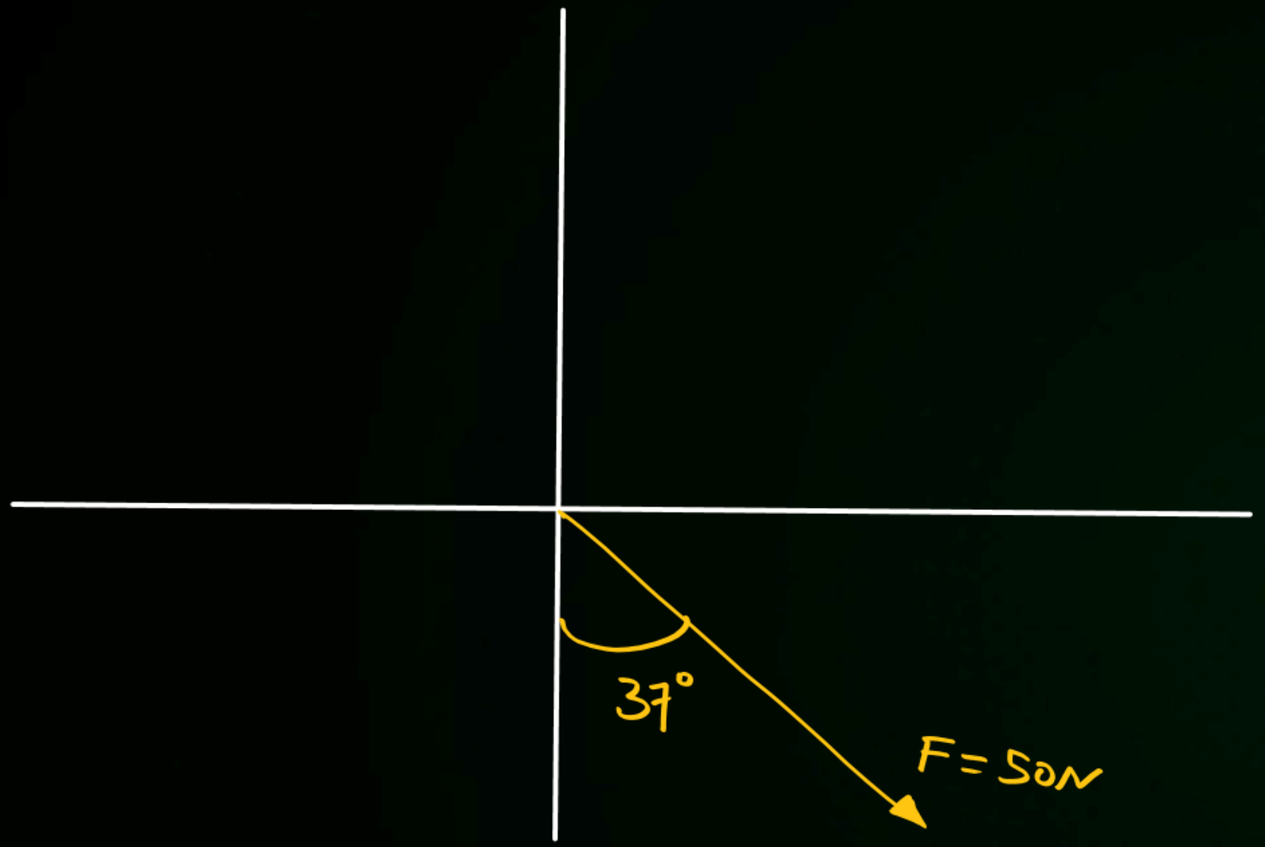
$\cos 37° = \frac{4}{5}$

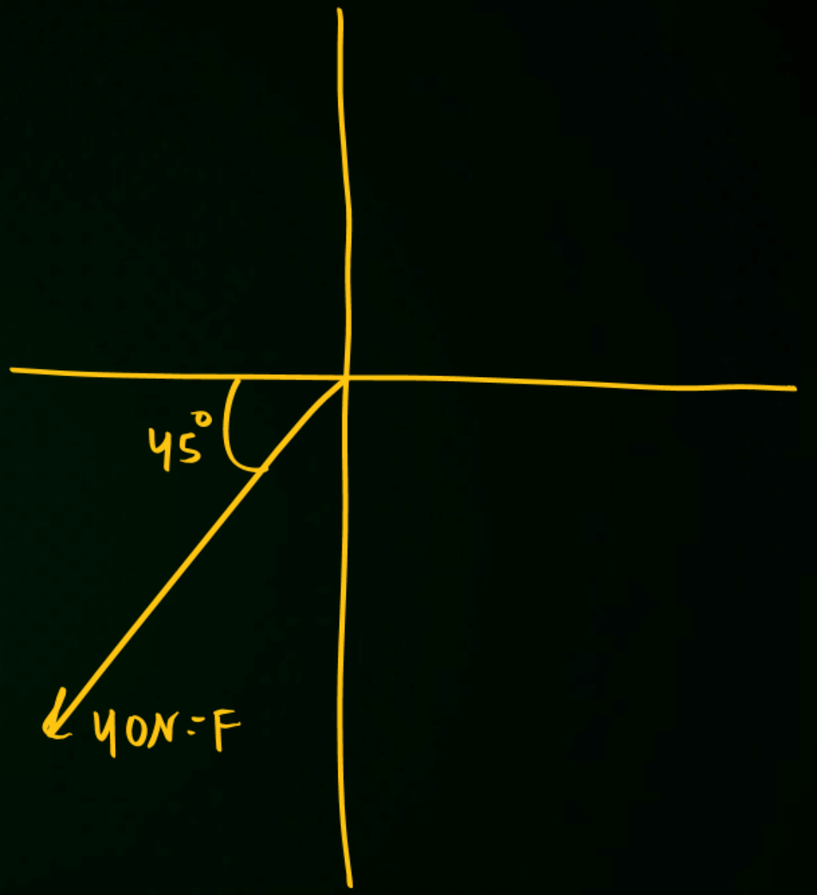
$\sin 37° = \frac{3}{5}$

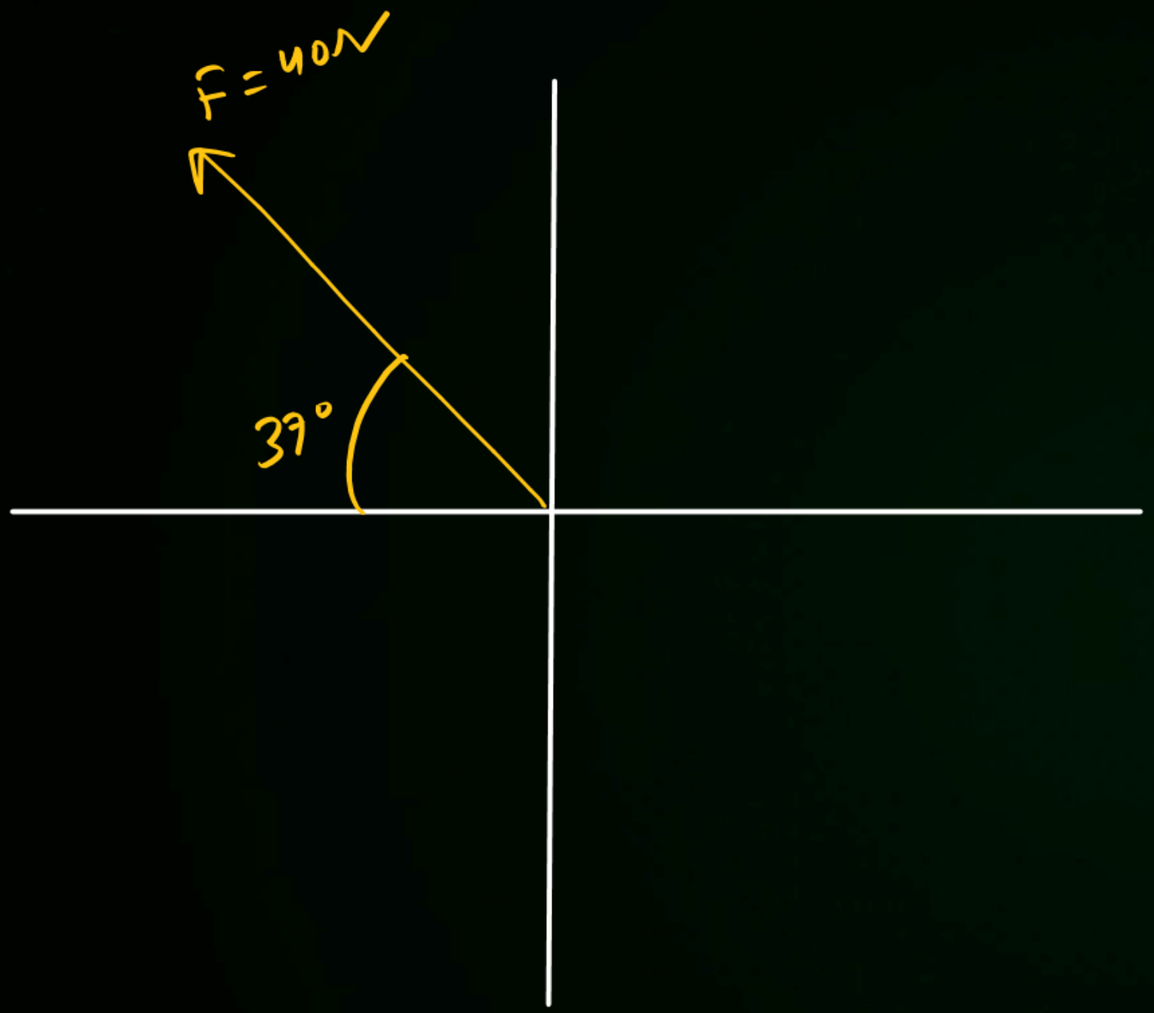
$\sin 53° = \frac{4}{5}$

$\cos 53° = \frac{3}{5}$

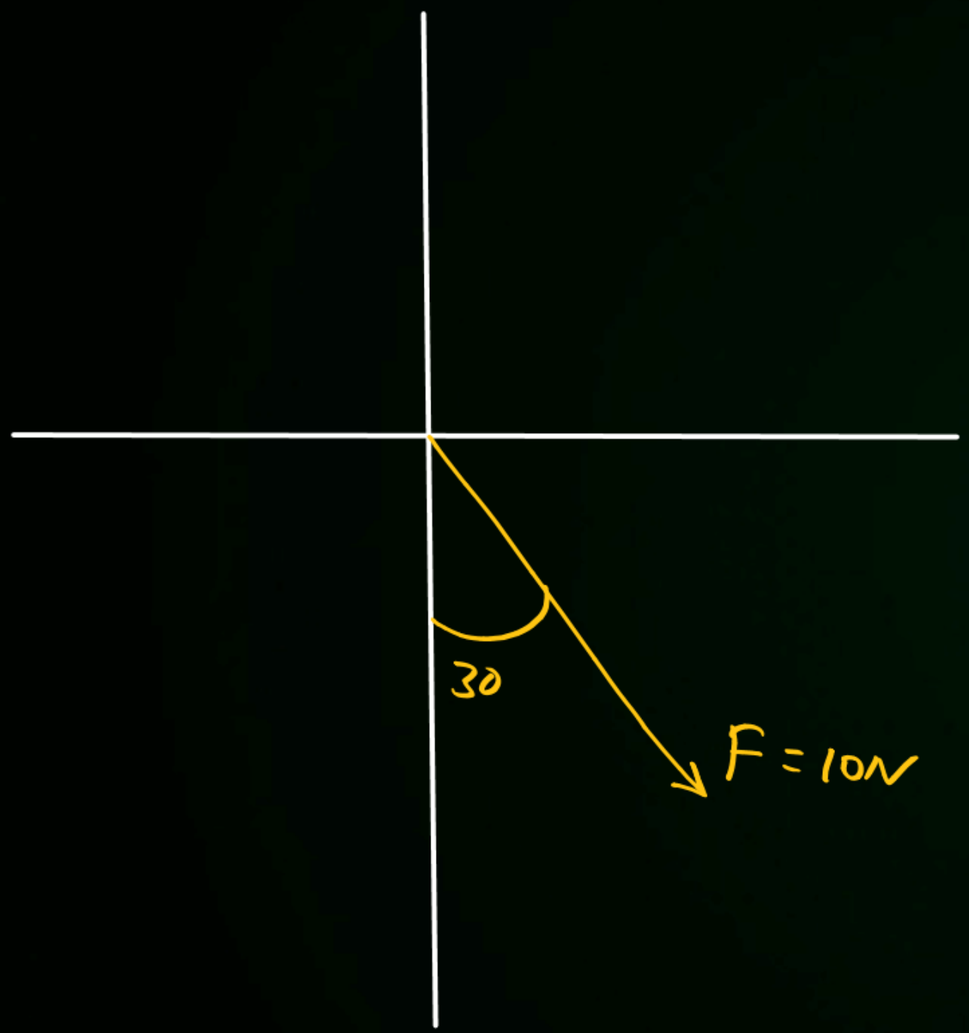
Take component






F = 40N
37°


F=50N
37°


30
F = 10N

Taking magnitude of vector

1. $\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$

$$|\vec{F}| = \sqrt{F_x^2 + F_y^2 + F_z^2}$$

(2) $\vec{F} = 30i + 40\hat{j}$

$$|\vec{F}| = \sqrt{(30)^2 + (40)^2}$$

$$= 50$$

3. $\vec{F} = 2\hat{i} + 2\hat{j}$

$$|F| = \sqrt{2^2 + 2^2} = \sqrt{4+4} = \sqrt{8} = \sqrt{4\times 2} = 2\sqrt{2}$$

4. $\vec{F} = 10\hat{i} + 10\hat{j}$

$|F| = 10\sqrt{2}$.

$\vec{F_1} = 3\hat{i} - 4\hat{j}$

$|\vec{F}| = \sqrt{(3)^2 + (-4)^2}$

$= \sqrt{9+16}$

$= \sqrt{25}$

$= 5$

$\vec{F_2} = -3\hat{i} + 4\hat{j}$

$\vec{F_3} = -3\hat{i} - 4\hat{j}$

$\vec{F_4} = 3\hat{i} + 4\hat{j}$

Question

H/W

Find magnitude of Vector:

$\vec{A} = 2\hat{i} + 3\hat{j}$ →

$\vec{B} = 3\hat{i} + 4\hat{j}$ →

$\vec{C} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ →

$\vec{D} = \hat{i} - \hat{j} + \hat{k}$ →

$\vec{E} = 6\hat{i} - 8\hat{j} + 10\hat{k}$ → $|E| = \sqrt{6^2 + (-8)^2 + (10)^2}$ ✓

$\vec{F} = 10\hat{i} - 10\hat{j} - 10\hat{k}$ →

Thank You